

GEOGRAPHY

INDIAN GEOGRAPHY



INDIAN SOILS

MODULE - 6

SOIL TYPES

MAJOR SOIL TYPES OF INDIA

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LATERITE & LATERITIC SOILS

WHAT DO YOU MEAN BY LATERISATION?

1. The term laterite is derived from the word later meaning brick or tile and was originally applied to a group of high clay Indian soils found in Malabar hills of Kerala, Tamil Nadu, Karnataka and Maharashtra.
2. It refers specifically to a particular cemented horizon in certain soils which when dried, become very hard, like a brick. Such soils (in tropics) when massively impregnated with **Sesquioxide (iron and aluminium oxides)** to extent of 70 to 80 per cent of the total mass, are called laterite soils. The soil forming process is called Laterization.
3. **Laterization is the process that removes silica, instead of sesquioxides from the upper layers and thereby leaving sesquioxides to concentrate in the solum.**

The process operates under the following conditions.

- i) **Climate**: The process of laterization operates most favorable in warm and humid (tropical) climate with 200-250 cm rainfall and continuous high temperature (25°C) throughout the year.
- ii) **Natural vegetation**: The rain forests of tropical areas are favorable for the process.
- iii) **Parent Material**: Basic parent materials, having sufficient iron bearing ferromagnesian minerals which on weathering release iron, are congenial for the development of laterites.

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LATERITE & LATERITIC SOILS



Gravel of red sand stones

- 1 Laterite soils are mostly the end products of weathering.
- 2 They are formed under conditions of high temperature and heavy rainfall with alternate wet and dry periods.
- 3 Heavy rainfall promotes leaching (nutrients gets washed away by water) of soil whereby lime and silica are leached away and a soil rich in oxides of iron and aluminium compounds is left behind.
- 4 'Laterite' means brick in Latin. They harden greatly on losing moisture.
- 5 Laterite soils are red in colour due to little clay and more gravel of red sand-stones.

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LATERITE & LATERITIC SOILS

DISTRIBUTION OF LATERITE SOILS



1 Laterite soils cover an area of 2.48 lakh sq km.

2 Continuous stretch of laterite soil is found on the summits of Western Ghats at 1000 to 1500 m above mean sea level, Eastern Ghats, the Rajmahal Hills, Vindhyan, Satpuras and Malwa Plateau.

3 They are well developed in south Maharashtra and parts of Karnataka, Andhra Pradesh, Orissa, West Bengal, Kerala, Assam and Meghalaya.

5. Laterite Soil



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LATERITE & LATERITIC SOILS

CROPS IN LATERITE SOILS



- 1 Laterite soils lack fertility due to intensive leaching.
- 2 When manured and irrigated, some laterites are suitable for growing plantation crops like arecanut, coffee, rubber, cinchona, coconut, tea, etc.
- 3 In some areas, these soils support grazing grounds and scrub forests.

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LATERITE & LATERITIC SOILS

ECONOMIC VALUE OF LATERITE SOILS



Laterite bricks

- 1 Laterite and lateritic soils have a unique distinction of providing valuable building material.
- 2 These soils can be easily cut with a spade but hardens like iron when exposed to air.
- 3 As it is the end-product of weathering, it cannot be weathered much further and is indefinitely durable.

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LATERITE & LATERITIC SOILS

CHEMICAL COMPOSITION



- 1 Laterite soils are rich in bauxite or ferric oxides.
- 2 They are very poor in lime, magnesia, potash and nitrogen.
- 3 Sometimes, the phosphate content may be high in the form of iron phosphate.
- 4 In wetter places, there may be higher content of humus.

MAJOR SOIL TYPES OF INDIA

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FOREST & MOUNTAIN SOILS

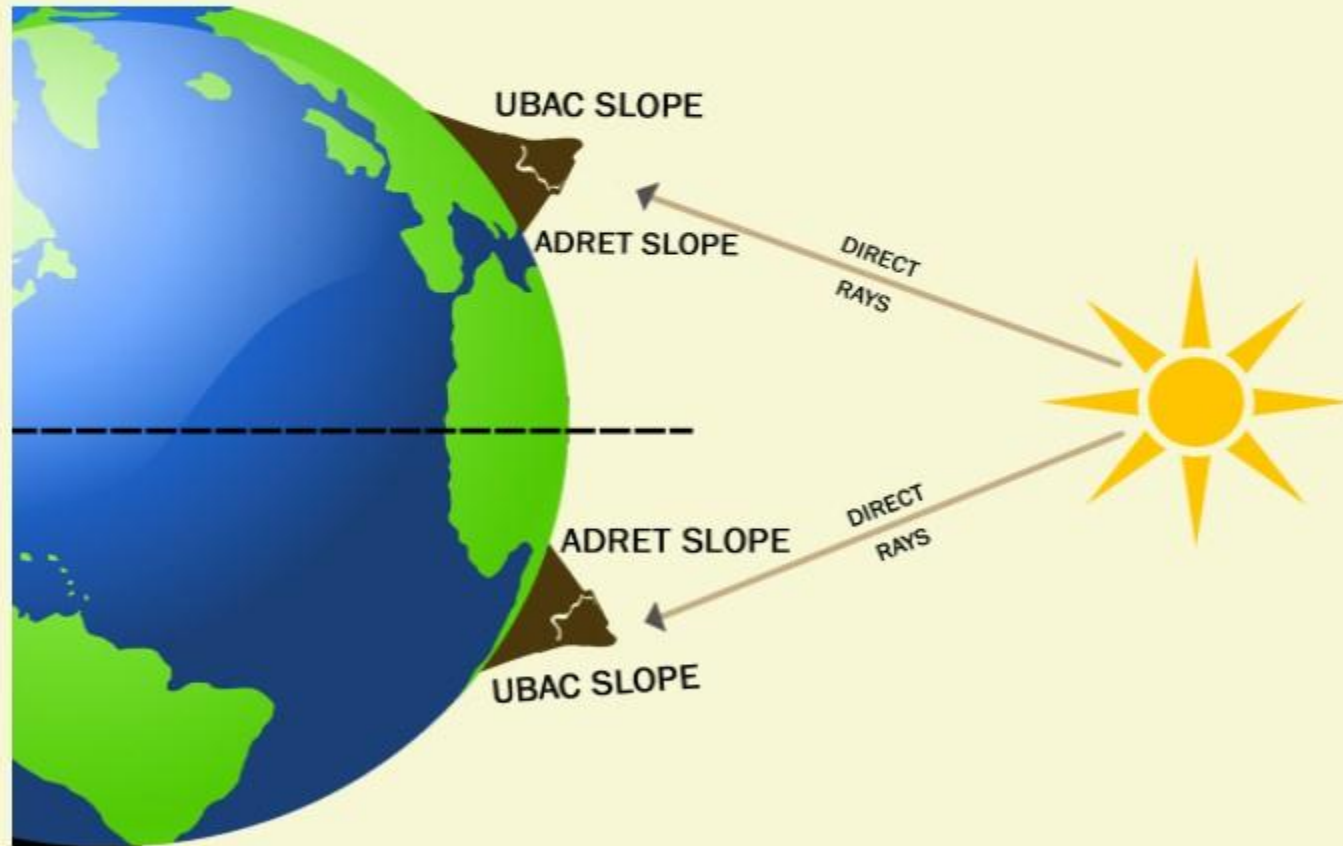
- 1 These soils occupy about 2.85 lakh sq km or 8.67% of the total land area of India.
- 2 They are mainly heterogeneous soils found on the hill slopes covered by forests. They are loamy and silty on valley sides and coarse-grained in the upper slopes.
- 3 In the snow-bound areas of the Himalayas, they experience denudation, and are acidic with low humus content. The soils found in the lower valleys are fertile.
- 4 The formation of these soils is mainly governed by the characteristic deposition of organic matter derived from forests and their character changes with parent rocks, ground-configuration and climate.
- 5 Consequently, they differ greatly even if they occur in close proximity to one another.

MAJOR SOIL TYPES OF INDIA

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FOREST & MOUNTAIN SOILS

DISTRIBUTION OF FOREST SOILS



1 In the Himalayan region, such soils are mainly found in valleys, less steeply inclined slopes.

2 Generally, it is the **north facing slopes** which support soil cover; the southern slopes being too precipitous and exposed to denudation and hence do not support soil formation.

3 Apart from the Himalayas, the forest soils occurs to some extent in Western and Eastern Ghats also.

1. Forest Soil of India



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FOREST & MOUNTAIN SOILS

CHEMICAL PROPERTIES



- 1 The forest soils are **very rich in humus**.
- 2 They are deficient in potash, phosphorus and lime.
- 3 They require good deal of fertilizers for high yields.

MAJOR SOIL TYPES OF INDIA

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FOREST & MOUNTAIN SOILS

CROPS IN FOREST SOILS



1

They are suitable for plantations of tea, coffee, spices and tropical fruits in Karnataka, Tamil Nadu and Kerala.

2

Wheat, maize, barley and temperate fruits are grown in Jammu and Kashmir, Himachal Pradesh and Uttarakhand.

MAJOR SOIL TYPES OF INDIA

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ARID – DESERT SOILS

- 1 The desert soils consist of Aeolian sand (90 to 95 per cent) and clay (5 to 10 per cent). They cover over 4.42% of the total area (1.42 lakh sq km) of the country.



- 2 These areas receive less than 50cm of rainfall and is affected by desert conditions.
- 3 This area is covered by a mantle of sand which inhibits plant growth.
- 4 This sand has originated from the mechanical disintegration of the ground rocks or is blown from the Indus basin by the prevailing south-west monsoon winds.
- 5 Expansion of deserts or desertification of neighboring soils is common due to intrusion of desert sand under the influence of wind [Aeolian sand].

6. Arid Soil



MAJOR SOIL TYPES OF INDIA

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ARID – DESERT SOILS

DISTRIBUTION OF ARID SOILS



1

Occur in arid and semi-arid regions of Rajasthan, west of the Aravallis, northern Gujarat, Saurashtra, Kutch, western parts of Haryana, and the south-western parts of Punjab.

2

The sand here is blown from the Indus basin and the coast by the prevailing south-west monsoon winds.

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ARID – DESERT SOILS

CHEMICAL PROPERTIES

- 1 They are usually poor in organic matter. Some desert soils are alkaline with varying degree of soluble salts like calcium carbonate. Over large areas, the calcium content increases downwards & in certain areas the subsoil has 10 times more calcium.
- 2 The 'Kankar' layer formation in the bottom horizons restricts the infiltration of water, and as such when irrigation is made available, the soil moisture is readily available for a sustainable plant growth.
- 3 The phosphate content of these soils is as high as in normal alluvial soils.
- 4 Nitrogen is originally low but its deficiency is made up to some extent by the availability of nitrogen in the form of nitrates.
- 5 Thus the presence of phosphates and nitrates makes them fertile soils wherever moisture is available.
- 6 There is, therefore, great possibility of reclaiming these soils if proper irrigation facilities are available.
- 7 The changes in the cropping pattern in the Indira Gandhi Canal Command Area is a living example of the utility of the desert soils.

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ARID – DESERT SOILS

CROPS IN ARID SOILS



- 1 Phosphates and nitrates make these soil fertile wherever moisture is available.
- 2 There is a possibility of reclaiming these soils if proper irrigation facilities are available.
- 3 In large areas of desert soils, only the drought resistant and salt tolerant crops such as barley, cotton, millets, maize and pulses are grown. Consequently, these soils support a low density of population.

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PEATY & MARSHY SOILS



- 1 These are soils with large amount of organic matter and considerable amount of soluble salts.
- 2 The most humid regions have this type of soil.
- 3 They are black, heavy and highly acidic.

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PEATY & MARSHY SOILS

DISTRIBUTION

- Kottayam and Alappuzha districts of Kerala where it is called kari.
- Also occur in the coastal areas of Odisha and Tamil Nadu, Sunderbans of West Bengal, in Bihar and Almora district of Uttarakhand.

CHEMICAL PROPERTIES

They are deficient in potash and phosphate.

CROPS OF PEATY SOILS

Most of the peaty soils are under water during the rainy season but as soon the rains cease, they are put under paddy cultivation.

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